

## Prevalence of zinc deficiency and its clinical relevance among hospitalised elderly

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### Abstract

Zinc is an essential trace element, and constituent of many metallo-enzymes required for normal metabolism. Age may be associated with altered metallothionein metabolism related to changes in zinc metabolism. The objectives of this study were: (i) to assess the prevalence of zinc deficiency among hospitalised elderly patients; (ii) to define the social, functional, pathological and nutritional characteristics of zinc deficient elderly hospitalised patients; and (iii) to assess the relationship between the zinc status and humoral immune function among hospitalised elderly patients. Fifty consecutive patients underwent comprehensive geriatric assessments included evaluations of the medical (index of the severity of the disease(s)), psychiatric (Geriatric depression scale (GDS)), therapeutic, social, functional (Katz's scale), and nutritional problems (Mini Nutritional Assessment (MNA) and biochemical markers (zinc, albumin, prealbumin (PAB), cholesterol) before their discharge. Fourteen patients (28%) presented a zinc concentrations lower than 10.7  $\mu\text{mol/l}$ , this value is usually considered

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